Developing Classification Models (Solution)

October 24, 2026

1 Import software libraries

[1]:
```
import sys                                                    # Read system␣
 ↪parameters.
import numpy as np                                            # Work with␣
 ↪multi-dimensional arrays.
import pandas as pd                                           # Manipulate and␣
 ↪analyze data.
import matplotlib                                             # Create and format␣
 ↪charts.
import matplotlib.pyplot as plt
import category_encoders as ce                                # Encode data.
import sklearn                                                # Train and␣
 ↪evaluate machine learning models.
from sklearn.model_selection import train_test_split, \
                                    learning_curve, \
                                    RandomizedSearchCV, \
                                    GridSearchCV
from sklearn.preprocessing import MinMaxScaler
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn import tree
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, \
                            confusion_matrix, \
                            f1_score, \
                            recall_score, \
                            precision_score, \
                            plot_roc_curve, \
                            plot_precision_recall_curve, \
                            plot_confusion_matrix
from sklearn.dummy import DummyClassifier
```

```
import xgboost                                          # Build gradient␣
 ↪boosting models.
from xgboost import XGBClassifier
import imblearn                                         # Deal with␣
 ↪imbalanced data.
from imblearn.over_sampling import SMOTE                # Perform␣
 ↪oversampling.
from collections import Counter                         # Count objects in␣
 ↪containers.
import pickle                                           # Save Python␣
 ↪objects as binary files.
import warnings                                         # Suppress warnings.
warnings.filterwarnings('ignore')

# Ensure results are reproducible.
np.random.seed(1)

# Summarize software libraries used.
print('Libraries used in this project:')
print('- Python {}'.format(sys.version))
print('- NumPy {}'.format(np.__version__))
print('- pandas {}'.format(pd.__version__))
print('- Matplotlib {}'.format(matplotlib.__version__))
print('- Category Encoders {}'.format(ce.__version__))
print('- scikit-learn {}'.format(sklearn.__version__))
print('- XGBoost {}'.format(xgboost.__version__))
print('- imbalanced-learn {}\n'.format(imblearn.__version__))
```

```
Libraries used in this project:
- Python 3.7.6 | packaged by conda-forge | (default, Mar 23 2020, 23:03:20)
[GCC 7.3.0]
- NumPy 1.19.2
- pandas 1.1.3
- Matplotlib 3.3.2
- Category Encoders 2.2.2
- scikit-learn 0.23.2
- XGBoost 1.3.3
- imbalanced-learn 0.7.0
```

2 Load and preview the data

[2]:
```
users_data = pd.read_pickle('/home/jovyan/work/Classification/data/
 ↪users_data_final.pickle')

users_data.head(n = 5)
```

[2]:

	user_id	number_transactions	total_amount_usd	job_management	job_technician	job_entrepreneur	job_blue-collar	job_retired	job_admin.	job_services	…	duration	campaign	pdays	previous	term_deposit	device_desktop	device_tablet	single	age_group_encoded	month_joined
0	9231c446-cb16-4b2b-a7f7-ddfc8b25aaf6	3.0	2143.00	1	0	0	0	0	0	0	…	261	1	-1	0	False	0	0	0	4	8
1	bb92765a-08de-4963-b432-496524b39157	0.0	1369.42	0	1	0	0	0	0	0	…	151	1	-1	0	False	1	0	1	2	7
2	573de577-49ef-42b9-83da-d3cfb817b5c1	2.0	2.00	0	0	1	0	0	0	0	…	76	1	-1	0	False	0	0	0	1	6
3	d6b66b9d-7c8f-4257-a682-e136f640b7e3	0.0	1369.42	0	0	0	1	0	0	0	…	92	1	-1	0	False	0	1	0	3	6
4	fade0b20-7594-4d9a-84cd-c02f79b1b526	1.0	1.00	0	0	0	0	0	0	0	…	198	1	-1	0	False	0	0	1	1	8

[5 rows x 33 columns]

3 Check the shape of the data

[3]:
```
users_data.shape
```

[3]:
```
(45179, 33)
```

4 Check the data types

[4]:
```
users_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 45179 entries, 0 to 45215
Data columns (total 33 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   user_id              45179 non-null  object
 1   number_transactions  45179 non-null  float64
 2   total_amount_usd     45179 non-null  float64
 3   job_management       45179 non-null  int64
 4   job_technician       45179 non-null  int64
 5   job_entrepreneur     45179 non-null  int64
 6   job_blue-collar      45179 non-null  int64
 7   job_retired          45179 non-null  int64
 8   job_admin.           45179 non-null  int64
 9   job_services         45179 non-null  int64
 10  job_self-employed    45179 non-null  int64
 11  job_unemployed       45179 non-null  int64
 12  job_housemaid        45179 non-null  int64
 13  job_student          45179 non-null  int64
 14  education_tertiary   45179 non-null  int64
 15  education_secondary  45179 non-null  int64
 16  education_Unknown    45179 non-null  int64
 17  education_primary    45179 non-null  int64
 18  default              45179 non-null  bool
 19  housing              45179 non-null  bool
 20  loan                 45179 non-null  bool
 21  contact_cellular     45179 non-null  int64
 22  contact_telephone    45179 non-null  int64
 23  duration             45179 non-null  int64
 24  campaign             45179 non-null  int64
 25  pdays                45179 non-null  int64
 26  previous             45179 non-null  int64
 27  term_deposit         45179 non-null  bool
 28  device_desktop       45179 non-null  int64
 29  device_tablet        45179 non-null  int64
```

```
 30  single                45179 non-null  uint8
 31  age_group_encoded     45179 non-null  int8
 32  month_joined          45179 non-null  int64
dtypes: bool(4), float64(2), int64(24), int8(1), object(1), uint8(1)
memory usage: 9.9+ MB
```

5 Explore the distribution of the target variable

[5]:
```
users_data.term_deposit.value_counts(normalize = True)
```

[5]:
```
False    0.883021
True     0.116979
Name: term_deposit, dtype: float64
```

6 Split the data into target and features

[6]:
```
target_data = users_data.term_deposit
features = users_data.drop(['user_id', 'term_deposit'], axis = 1)
```

7 Split the data into train and test sets

[7]:
```
X_train, X_test, y_train, y_test = train_test_split(features,
                                                    target_data,
                                                    test_size = 0.3)
```

[8]:
```
print('Training data features: ', X_train.shape)
print('Training data target:   ', y_train.shape)
```

```
Training data features:  (31625, 31)
Training data target:    (31625,)
```

8 Apply oversampling to the data

[9]:
```
print('Before oversampling: ', Counter(y_train))
```

```
Before oversampling:  Counter({False: 27963, True: 3662})
```

[10]:
```
# Define oversampling strategy.

SMOTE = SMOTE()
```

```
# Fit and apply the transform.

X_train_SMOTE, y_train_SMOTE = SMOTE.fit_resample(X_train, y_train)
X_train_SMOTE = pd.DataFrame(X_train_SMOTE,
                             columns = X_train.columns)

print('After oversampling: ', Counter(y_train_SMOTE))
```

```
After oversampling:  Counter({False: 27963, True: 27963})
```

9 Check the distribution of the test data

[11]:
```
# Test data should not be oversampled.

print('Test data features: ', X_test.shape)
print('Test data target:   ', y_test.shape)
```

```
Test data features:  (13554, 31)
Test data target:    (13554,)
```

[12]:
```
Counter(y_test)
```

[12]:
```
Counter({False: 11931, True: 1623})
```

10 Normalize the data

[13]:
```
norm = MinMaxScaler().fit(X_train_SMOTE)
```

[14]:
```
X_train_norm = norm.transform(X_train_SMOTE)

print('Minimum: ', np.min(X_train_norm))
print('Maximum: ', np.max(X_train_norm))
```

```
Minimum:  0.0
Maximum:  1.0
```

11 Train a logistic regression model

[15]:
```
logreg = LogisticRegression()
logreg.fit(X_train_norm, y_train_SMOTE)
```

[15]:
```
LogisticRegression()
```

12 Make predictions using the logistic regression model

[16]:
```
logreg_y_pred = logreg.predict(X_test)
print(Counter(logreg_y_pred))
```

```
Counter({True: 13486, False: 68})
```

[17]:
```
results = pd.concat([y_test.iloc[:5], X_test.iloc[:5]], axis = 1)
results.insert(1, 'term_deposit_pred', logreg_y_pred[:5])
results
```

[17]:

	term_deposit	term_deposit_pred	number_transactions	total_amount_usd	\
43308	False	True	0.0	1369.42	
32770	False	True	2.0	246.00	
17440	False	True	0.0	1369.42	
36164	False	True	0.0	1369.42	
29218	False	True	0.0	1369.42	

	job_management	job_technician	job_entrepreneur	job_blue-collar	\
43308	1	0	0	0	
32770	0	0	0	1	
17440	0	0	0	0	
36164	0	0	0	1	
29218	0	0	0	0	

	job_retired	job_admin.	…	contact_telephone	duration	campaign	\
43308	0	0	…	1	129	1	
32770	0	0	…	0	191	1	
17440	0	0	…	0	780	2	
36164	0	0	…	0	114	2	
29218	0	0	…	0	111	1	

	pdays	previous	device_desktop	device_tablet	single	\
43308	188	2	0	0	0	
32770	150	7	0	0	0	
17440	-1	0	0	0	0	
36164	-1	0	1	0	1	
29218	-1	0	0	0	0	

	age_group_encoded	month_joined
43308	2	3
32770	1	7
17440	2	5

```
36164                  2            10
29218                  2             5

[5 rows x 33 columns]
```

13 Obtain the logistic regression model's score

[18]:
```
accuracy_score(y_test, logreg_y_pred)
```

[18]:
```
0.1247602183857164
```

14 Train a *k*-nearest neighbor (*k*-NN) model

[19]:
```
knn = KNeighborsClassifier()
knn.fit(X_train_norm, y_train_SMOTE)
```

[19]:
```
KNeighborsClassifier()
```

15 Make predictions using the *k*-NN model

[20]:
```
knn_y_pred = knn.predict(X_test)
print(Counter(knn_y_pred))
```

```
Counter({True: 10197, False: 3357})
```

[21]:
```
results['term_deposit_pred'] = knn_y_pred[:5]
results
```

[21]:
```
       term_deposit  term_deposit_pred  number_transactions  total_amount_usd  \
43308         False               True                  0.0           1369.42
32770         False               True                  2.0            246.00
17440         False               True                  0.0           1369.42
36164         False               True                  0.0           1369.42
29218         False               True                  0.0           1369.42

       job_management  job_technician  job_entrepreneur  job_blue-collar  \
43308               1               0                 0                0
32770               0               0                 0                1
17440               0               0                 0                0
36164               0               0                 0                1
29218               0               0                 0                0
```

```
       job_retired  job_admin.  …  contact_telephone  duration  campaign  \
43308            0           0  …                  1       129         1
32770            0           0  …                  0       191         1
17440            0           0  …                  0       780         2
36164            0           0  …                  0       114         2
29218            0           0  …                  0       111         1

       pdays  previous  device_desktop  device_tablet  single  \
43308    188         2               0              0       0
32770    150         7               0              0       0
17440     -1         0               0              0       0
36164     -1         0               1              0       1
29218     -1         0               0              0       0

       age_group_encoded  month_joined
43308                  2             3
32770                  1             7
17440                  2             5
36164                  2            10
29218                  2             5

[5 rows x 33 columns]
```

16 Obtain the *k*-NN model's score

[22]:
```
accuracy_score(y_test, knn_y_pred)
```

[22]:
```
0.2968865279622252
```

[23]:
```
print('Number of mislabeled points out of a total %d points: %d'
      % (X_test.shape[0], (y_test != knn_y_pred).sum()))
```

```
Number of mislabeled points out of a total 13554 points: 9530
```

17 Train a support-vector machine (SVM) model

[24]:
```
svm = SVC()
svm.fit(X_train_norm, y_train_SMOTE)
```

[24]:
```
SVC()
```

18 Make predictions using the SVM model

[25]:
```
svm_y_pred = svm.predict(X_test)
print(Counter(svm_y_pred))
```

```
Counter({False: 13554})
```

[26]:
```
results['term_deposit_pred'] = svm_y_pred[:5]
results
```

[26]:

	term_deposit	term_deposit_pred	number_transactions	total_amount_usd	\
43308	False	False	0.0	1369.42	
32770	False	False	2.0	246.00	
17440	False	False	0.0	1369.42	
36164	False	False	0.0	1369.42	
29218	False	False	0.0	1369.42	

	job_management	job_technician	job_entrepreneur	job_blue-collar	\
43308	1	0	0	0	
32770	0	0	0	1	
17440	0	0	0	0	
36164	0	0	0	1	
29218	0	0	0	0	

	job_retired	job_admin.	…	contact_telephone	duration	campaign	\
43308	0	0	…	1	129	1	
32770	0	0	…	0	191	1	
17440	0	0	…	0	780	2	
36164	0	0	…	0	114	2	
29218	0	0	…	0	111	1	

	pdays	previous	device_desktop	device_tablet	single	\
43308	188	2	0	0	0	
32770	150	7	0	0	0	
17440	-1	0	0	0	0	
36164	-1	0	1	0	1	
29218	-1	0	0	0	0	

	age_group_encoded	month_joined
43308	2	3
32770	1	7
17440	2	5
36164	2	10
29218	2	5

[5 rows x 33 columns]

19 Obtain the SVM model's score

[27]:
```
accuracy_score(y_test, svm_y_pred)
```

[27]:
```
0.880256750774679
```

20 Train a naïve Bayes model

[28]:
```
gnb = GaussianNB()
gnb.fit(X_train_SMOTE, y_train_SMOTE)
```

[28]:
```
GaussianNB()
```

21 Make predictions using the naïve Bayes model

[29]:
```
gnb_y_pred = gnb.predict(X_test)
print(Counter(gnb_y_pred))
```

```
Counter({False: 9685, True: 3869})
```

[30]:
```
results['term_deposit_pred'] = gnb_y_pred[:5]
results
```

[30]:

	term_deposit	term_deposit_pred	number_transactions	total_amount_usd	\
43308	False	False	0.0	1369.42	
32770	False	True	2.0	246.00	
17440	False	False	0.0	1369.42	
36164	False	False	0.0	1369.42	
29218	False	False	0.0	1369.42	

	job_management	job_technician	job_entrepreneur	job_blue-collar	\
43308	1	0	0	0	
32770	0	0	0	1	
17440	0	0	0	0	
36164	0	0	0	1	
29218	0	0	0	0	

	job_retired	job_admin.	…	contact_telephone	duration	campaign	\
43308	0	0	…	1	129	1	
32770	0	0	…	0	191	1	
17440	0	0	…	0	780	2	
36164	0	0	…	0	114	2	
29218	0	0	…	0	111	1	

```
       pdays  previous  device_desktop  device_tablet  single  \
43308    188         2               0              0       0
32770    150         7               0              0       0
17440     -1         0               0              0       0
36164     -1         0               1              0       1
29218     -1         0               0              0       0

       age_group_encoded  month_joined
43308                  2             3
32770                  1             7
17440                  2             5
36164                  2            10
29218                  2             5

[5 rows x 33 columns]
```

22 Obtain the naïve Bayes model's score

[31]:
```
accuracy_score(y_test, gnb_y_pred)
```

[31]:
```
0.7019330087059171
```

23 Train a decision tree model

[32]:
```
clf_tree = DecisionTreeClassifier()
clf_tree.fit(X_train_SMOTE, y_train_SMOTE)
```

[32]:
```
DecisionTreeClassifier()
```

24 Make predictions using the decision tree model

[33]:
```
clf_tree_y_pred = clf_tree.predict(X_test)
print(Counter(clf_tree_y_pred))
```

```
Counter({False: 11596, True: 1958})
```

[34]:
```
results['term_deposit_pred'] = clf_tree_y_pred[:5]
results
```

[34]:
```
       term_deposit  term_deposit_pred  number_transactions  total_amount_usd  \
43308         False               True                  0.0           1369.42
32770         False              False                  2.0            246.00
17440         False               True                  0.0           1369.42
36164         False              False                  0.0           1369.42
29218         False              False                  0.0           1369.42

       job_management  job_technician  job_entrepreneur  job_blue-collar  \
43308               1               0                 0                0
32770               0               0                 0                1
17440               0               0                 0                0
36164               0               0                 0                1
29218               0               0                 0                0

       job_retired  job_admin.  …  contact_telephone  duration  campaign  \
43308            0           0  …                  1       129         1
32770            0           0  …                  0       191         1
17440            0           0  …                  0       780         2
36164            0           0  …                  0       114         2
29218            0           0  …                  0       111         1

       pdays  previous  device_desktop  device_tablet  single  \
43308    188         2               0              0       0
32770    150         7               0              0       0
17440     -1         0               0              0       0
36164     -1         0               1              0       1
29218     -1         0               0              0       0

       age_group_encoded  month_joined
43308                  2             3
32770                  1             7
17440                  2             5
36164                  2            10
29218                  2             5

[5 rows x 33 columns]
```

25 Obtain the decision tree model's score

[35]:
```
accuracy_score(y_test, clf_tree_y_pred)
```

[35]:
```
0.8314150804190645
```

26 Visualize the decision tree

[36]:
```
text_representation = tree.export_text(clf_tree)
print(text_representation)
```

```
|--- feature_22 <= 201.50
|   |--- feature_22 <= 114.50
|   |   |--- feature_22 <= 77.50
|   |   |   |--- feature_24 <= 372.50
|   |   |   |   |--- feature_23 <= 3.50
|   |   |   |   |   |--- feature_25 <= 3.50
|   |   |   |   |   |   |--- feature_1 <= 10271.10
|   |   |   |   |   |   |   |--- feature_26 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 320.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  320.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_30 >  1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_26 >  0.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_1 >  10271.10
|   |   |   |   |   |   |   |   |--- feature_1 <= 10355.84
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_1 >  10355.84
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_25 >  3.50
|   |   |   |   |   |   |--- feature_1 <= -9.89
|   |   |   |   |   |   |   |--- feature_1 <= -64.76
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_1 >  -64.76
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_1 >  -9.89
|   |   |   |   |   |   |   |--- feature_18 <= 0.50
```

```
|   |   |   |   |   |   |   |   |--- feature_23 <= 2.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 2446.66
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  2446.66
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 66.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  66.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_23 >  2.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 7.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  7.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_12 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_12 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 296.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_1 >  296.00
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_23 >  3.50
|   |   |   |   |   |   |--- feature_25 <= 4.50
|   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_25 >  4.50
|   |   |   |   |   |   |   |--- feature_1 <= -16.00
|   |   |   |   |   |   |   |   |--- feature_22 <= 48.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_22 >  48.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_1 >  -16.00
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |--- feature_24 >  372.50
|   |   |   |   |   |--- feature_1 <= 53.00
|   |   |   |   |   |   |--- feature_1 <= 10.16
|   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_1 >  10.16
|   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |--- feature_1 >  53.00
|   |   |   |   |   |   |--- feature_22 <= 60.50
|   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_22 >  60.50
|   |   |   |   |   |   |   |--- feature_24 <= 387.50
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_24 >  387.50
|   |   |   |   |   |   |   |--- class: False
|   |   |   |--- feature_22 >  77.50
|   |   |   |   |--- feature_14 <= 0.50
```

```
|   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |--- feature_25 <= 7.50
|   |   |   |   |   |   |   |   |--- feature_24 <= 182.00
|   |   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_24 >  182.00
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 184.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  184.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- feature_25 >  7.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |--- feature_12 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_30 <= 1.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 82.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1348.21
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1348.21
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 79.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  79.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_22 >  82.50
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 5.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  5.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_30 >  1.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 4632.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 112.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_7 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_7 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  112.50
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 138.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  138.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_1 >  4632.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 4676.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  4676.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_12 >  0.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 102.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  102.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |--- feature_24 <= 64.50
|   |   |   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_0 <= 2.31
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_0 >  2.31
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.01
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 82.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 7.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  7.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  82.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 109.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  109.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |--- feature_0 >  2.01
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.97
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.97
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.68
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.68
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 30.00
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1.50
```

```
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  30.00
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 94.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  94.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |--- feature_24 >  64.50
|   |   |   |   |   |   |   |--- feature_24 <= 97.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 194.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_1 >  194.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 82.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  82.50
|   |   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 581.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  581.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |--- feature_24 >  97.50
|   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_25 <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 110.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  110.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_25 >  1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 181.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  181.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_26 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |   |   |   |--- feature_26 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_25 <= 10.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 1.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  1.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_25 >  10.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 11.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  11.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |--- feature_14 >  0.50
|   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |--- feature_0 <= 1.00
|   |   |   |   |   |   |   |--- feature_22 <= 94.00
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_22 >  94.00
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_0 >  1.00
|   |   |   |   |   |   |   |--- feature_24 <= 140.00
|   |   |   |   |   |   |   |   |--- feature_1 <= 2936.00
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_1 >  2936.00
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_24 >  140.00
|   |   |   |   |   |   |   |   |--- feature_30 <= 7.00
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_30 >  7.00
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |--- feature_1 <= 2924.15
|   |   |   |   |   |   |   |--- feature_25 <= 1.50
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_25 >  1.50
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_1 >  2924.15
```

```
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |--- feature_0 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 100.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  100.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 108.50
|   |   |   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  108.50
|   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_0 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.49
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 79.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_22 >  79.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_0 >  2.49
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 3.94
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  3.94
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 608.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  608.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |--- feature_24 <= 350.00
|   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 111.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1484.61
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1484.61
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  111.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
```

```
|   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |--- feature_24 <= 183.00
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1015.07
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1015.07
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |--- feature_24 >  183.00
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.41
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.41
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- feature_24 >  350.00
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |--- feature_24 <= 382.50
|   |   |   |   |   |   |   |--- feature_0 <= 2.14
|   |   |   |   |   |   |   |   |--- feature_22 <= 88.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= -19.94
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  -19.94
|   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |--- feature_22 >  88.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 1654.14
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1654.14
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1690.14
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1690.14
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_0 >  2.14
|   |   |   |   |   |   |   |   |--- feature_0 <= 2.97
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_0 >  2.97
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 158.24
|   |   |   |   |   |   |   |   |   |   |--- feature_27 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_27 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_1 >  158.24
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 160.24
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  160.24
```

```
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |--- feature_24 >  382.50
|   |   |   |   |   |   |   |   |--- class: True
|   |--- feature_22 >  114.50
|   |   |--- feature_24 <= -0.50
|   |   |   |--- feature_23 <= 1.50
|   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_12 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- feature_12 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 749.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  749.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_26 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 66.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_11 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_11 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  66.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_9 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 16
|   |   |   |   |   |   |   |   |   |   |   |--- feature_9 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- feature_26 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 321.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  321.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 104.50
|   |   |   |   |   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.31
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
```

```
|   |   |   |   |   |   |   |   |   |--- feature_0 >  2.31
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_4 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_4 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_1 >  104.50
|   |   |   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_5 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 13
|   |   |   |   |   |   |   |   |   |   |   |--- feature_5 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 223.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 137.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  137.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_1 >  223.00
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 7542.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  7542.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |--- feature_29 <= 4.50
|   |   |   |   |   |   |   |--- feature_30 <= 11.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 3642.00
|   |   |   |   |   |   |   |   |   |--- feature_10 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_10 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_1 >  3642.00
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 8.00
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  8.00
```

```
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |---  feature_30 >  11.50
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 12.50
|   |   |   |   |   |   |   |   |   |   |--- feature_21 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_21 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_5 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_5 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  12.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |---  feature_29 >  4.50
|   |   |   |   |   |   |   |   |   |--- feature_21 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 152.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_22 >  152.00
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 5.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_29 >  5.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 174.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  174.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |---  feature_21 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 9.00
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  9.00
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 137.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  137.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 176.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  176.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |---  feature_18 >  0.50
|   |   |   |   |   |--- feature_5 <= 0.50
|   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_8 >  0.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 176.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  176.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 339.61
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 115.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  115.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_21 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_21 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_1 >  339.61
|   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 200.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  200.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_7 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_7 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 120.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  120.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 16.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1957.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1957.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  16.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 17.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  17.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
```

```
|   |   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 1478.99
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1478.99
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1562.27
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1562.27
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_5 >  0.50
|   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.38
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1672.84
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1672.84
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_0 >  3.38
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.88
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_0 >  3.88
|   |   |   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 100.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  100.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |--- feature_22 <= 182.00
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_22 >  182.00
|   |   |   |   |   |   |   |   |--- feature_30 <= 7.50
|   |   |   |   |   |   |   |   |   |--- class: False
```

```
|   |   |   |   |   |   |   |   |--- feature_30 >  7.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |--- feature_1 <= 186.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 0.22
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 122.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_22 >  122.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_1 >  0.22
|   |   |   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_1 >  186.50
|   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 394.07
|   |   |   |   |   |   |   |   |   |   |   |--- feature_27 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_27 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  394.07
|   |   |   |   |   |   |   |   |   |   |   |--- feature_12 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_12 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |--- feature_23 <= 3.50
|   |   |   |   |   |   |   |--- feature_5 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_29 <= 4.50
|   |   |   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |   |   |--- feature_2 >  0.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 13
|   |   |   |   |   |   |   |   |--- feature_29 >  4.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 6.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1327.21
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1327.21
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.31
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.31
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_30 >  6.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 127.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  127.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 179.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  179.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |--- feature_5 >  0.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 1689.50
|   |   |   |   |   |   |   |   |   |--- feature_27 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 181.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  181.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 182.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  182.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_27 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 845.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  845.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_30 >  3.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_1 >  1689.50
```

```
|   |   |   |   |   |   |   |   |--- feature_1 <= 1781.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 176.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_22 >  176.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_1 >  1781.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_23 >  3.50
|   |   |   |   |   |   |   |--- feature_12 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.46
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 4815.97
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  4815.97
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.46
|   |   |   |   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 4273.12
|   |   |   |   |   |   |   |   |   |   |   |--- feature_10 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_10 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  4273.12
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 4639.74
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  4639.74
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 186.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 17925.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  17925.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 18727.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  18727.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  186.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 4.30
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 3586.79
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  3586.79
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_0 >  4.30
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 4.89
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  4.89
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |--- feature_12 >  0.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 2052.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 3.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_30 >  3.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 11.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  11.00
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_1 >  2052.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |--- feature_24 >  -0.50
|   |   |   |   |--- feature_24 <= 198.50
|   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 167.50
|   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 8.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_25 >  8.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 3.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  3.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_22 >  167.50
|   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 9.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_25 >  9.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |--- feature_0 <= 2.02
|   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 9.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1126.30
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1126.30
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |--- feature_30 >  9.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 97.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  97.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 114.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 87.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  87.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  114.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 196.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  196.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_0 >  2.02
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.00
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_0 >  3.00
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 151.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 14
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  151.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |--- feature_24 <= 106.00
|   |   |   |   |   |   |   |   |--- feature_25 <= 3.50
|   |   |   |   |   |   |   |   |   |--- feature_5 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 190.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 5.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  5.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  190.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_5 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_25 >  3.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_24 >  106.00
```

```
|   |   |   |   |   |   |   |   |--- feature_22 <= 164.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  164.50
|   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 5.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 196.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  196.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  5.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_24 <= 101.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 139.50
|   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_25 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_22 >  139.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 10.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  10.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_24 >  101.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 177.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_25 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 126.50
|   |   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  126.50
|   |   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
```

```
|   |   |   |   |   |   |   |   |--- feature_24 >  177.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 10.50
|   |   |   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  10.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 147.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  147.50
|   |   |   |   |   |   |   |   |   |--- feature_24 <= 97.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 182.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  182.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_24 >  97.50
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 193.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  193.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |--- feature_22 <= 168.50
|   |   |   |   |   |   |   |   |--- feature_24 <= 110.50
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 1.00
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 28.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  28.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_26 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_26 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_0 >  1.00
|   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= -23.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  -23.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_24 >  110.50
|   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_12 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_12 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 147.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  147.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 155.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  155.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_22 >  168.50
|   |   |   |   |   |   |   |   |--- feature_30 <= 8.50
|   |   |   |   |   |   |   |   |   |--- feature_24 <= 177.00
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 190.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  190.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_24 >  177.00
|   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 3.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  3.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- feature_30 >  8.50
|   |   |   |   |   |   |   |   |   |--- feature_11 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_11 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |--- feature_24 >  198.50
|   |   |   |   |   |--- feature_5 <= 0.50
|   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 1963.21
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1963.21
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 2662.42
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  2662.42
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |--- feature_25 <= 2.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 169.50
|   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 2433.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  2433.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 148.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  148.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |--- feature_22 >  169.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 260.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 181.56
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  181.56
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  260.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 172.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  172.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |--- feature_25 >  2.50
|   |   |   |   |   |   |   |   |   |--- feature_24 <= 374.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 271.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 267.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  267.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_24 >  271.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_24 >  374.50
```

```
|   |   |   |   |   |   |   |   |   |--- feature_24 <= 442.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_24 >  442.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 4.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  4.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |--- feature_24 <= 375.00
|   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |--- feature_24 <= 239.50
|   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 162.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  162.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_24 >  239.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.07
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1.89
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1.89
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.07
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 262.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  262.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 1034.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1034.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1135.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1135.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.12
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.12
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |--- feature_24 >  375.00
|   |   |   |   |   |   |   |   |--- feature_23 <= 2.50
```

```
|   |   |   |   |   |   |   |   |--- feature_1 <= 3234.30
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 384.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_1 >  384.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 931.48
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  931.48
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_1 >  3234.30
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_23 >  2.50
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_5 >  0.50
|   |   |   |   |   |   |--- feature_1 <= 342.50
|   |   |   |   |   |   |   |--- feature_1 <= 300.45
|   |   |   |   |   |   |   |   |--- feature_29 <= 3.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_29 >  3.50
|   |   |   |   |   |   |   |   |   |--- feature_24 <= 316.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_24 >  316.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_1 >  300.45
|   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_1 >  342.50
|   |   |   |   |   |   |   |--- class: False
|---  feature_22 >  201.50
|   |--- feature_22 <= 466.50
|   |   |--- feature_24 <= -0.50
|   |   |   |--- feature_5 <= 0.50
|   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 1475.00
|   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 266.00
|   |   |   |   |   |   |   |   |   |   |--- class: False
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  266.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1475.00
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 2254.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_4 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_4 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  2254.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |--- feature_29 <= 4.50
|   |   |   |   |   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 258.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  258.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 10820.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 454.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  454.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  10820.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 12190.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  12190.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_29 >  4.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 7.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 370.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.13
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.13
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  370.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_30 >  7.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1836.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 228.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  228.00
```

```
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1836.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 6552.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  6552.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_4 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_9 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_7 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 14
|   |   |   |   |   |   |   |   |   |   |--- feature_7 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |   |--- feature_9 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |--- feature_4 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 411.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  411.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 6517.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 16
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  6517.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 231.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  231.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 18
|   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 11794.47
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 192.17
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  192.17
```

```
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  11794.47
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 13078.47
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  13078.47
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |--- feature_29 <= 4.50
|   |   |   |   |   |   |   |   |   |--- feature_21 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 413.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  413.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 228.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  228.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 14
|   |   |   |   |   |   |   |   |   |--- feature_21 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1929.21
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1929.21
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 217.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  217.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_29 >  4.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_7 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_10 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 13
|   |   |   |   |   |   |   |   |   |   |--- feature_10 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 444.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  444.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_7 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 227.50
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |   |--- feature_22 >  227.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 341.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  341.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 193.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_21 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_21 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  193.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.34
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.34
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 434.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1364.21
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1364.21
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  434.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 322.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 1858.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 209.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  209.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1858.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 2016.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  2016.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
```

```
|   |   |   |   |   |   |   |   |--- feature_22 >  322.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |--- feature_22 <= 394.50
|   |   |   |   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 210.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  210.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 11.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  11.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.23
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.23
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 656.47
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  656.47
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_26 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 14
|   |   |   |   |   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_26 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 3250.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  3250.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 370.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  370.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 340.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 13
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  340.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |--- feature_22 >  394.50
|   |   |   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_7 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_11 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |   |   |   |--- feature_11 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 5.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  5.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_7 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  2.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= -660.00
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_1 >  -660.00
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 702.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  702.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |--- feature_5 >  0.50
|   |   |   |   |   |--- feature_22 <= 455.50
```

```
|   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 990.21
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 7.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  7.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  990.21
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 3584.76
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 9.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  9.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_1 >  3584.76
|   |   |   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 379.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 11.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 4720.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  4720.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_30 >  11.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 1.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  1.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  379.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 383.00
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  383.00
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 799.10
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  799.10
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |--- feature_1 <= -518.50
|   |   |   |   |   |   |   |   |--- feature_1 <= -538.00
```

```
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  -538.00
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_1 >  -518.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 254.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 3.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 246.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  246.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 250.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  250.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  3.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  254.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 1097.00
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 26.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  26.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1097.00
|   |   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 3551.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  3551.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 388.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 14369.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 6.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_13 >  0.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  6.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 232.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  232.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |   |--- feature_1 >  14369.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 15412.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  15412.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  388.50
|   |   |   |   |   |   |   |   |   |--- feature_29 <= 3.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 557.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  557.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 391.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  391.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- feature_29 >  3.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 1.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  1.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |--- feature_22 >  455.50
|   |   |   |   |   |--- feature_30 <= 3.50
|   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |--- feature_1 <= 8585.21
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_1 >  8585.21
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_30 >  3.50
|   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |--- feature_23 <= 2.50
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_23 >  2.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 83.21
```

```
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_1 >  83.21
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |--- feature_24 >  -0.50
|   |   |   |--- feature_5 <= 0.50
|   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |--- feature_7 <= 0.50
|   |   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_4 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_9 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_10 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 15
|   |   |   |   |   |   |   |   |   |   |--- feature_10 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |--- feature_9 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |--- feature_4 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 97.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 842.21
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  842.21
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  97.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 422.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  422.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_26 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.99
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 15
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.99
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 106.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  106.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |   |--- feature_26 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 100.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 42.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  42.00
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  100.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 9.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 181.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 3.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_25 >  3.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  181.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 99.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  99.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |--- feature_30 >  9.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 349.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 210.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  210.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  349.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_24 <= 93.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 215.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  215.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_24 >  93.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 370.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 325.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  325.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_24 >  370.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_7 >  0.50
```

```
|   |   |   |   |   |   |   |--- feature_24 <= 116.00
|   |   |   |   |   |   |   |   |--- feature_24 <= 49.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_24 >  49.50
|   |   |   |   |   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 9.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  9.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_27 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_27 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_25 >  4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_24 >  116.00
|   |   |   |   |   |   |   |   |--- feature_24 <= 375.00
|   |   |   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 452.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  452.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 193.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  193.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 363.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  363.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |--- feature_24 >  375.00
|   |   |   |   |   |   |   |   |--- feature_22 <= 252.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  252.50
|   |   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 350.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  350.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |--- feature_18 <= 0.50
|   |   |   |   |   |   |--- feature_22 <= 235.50
|   |   |   |   |   |   |   |--- feature_24 <= 426.00
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_24 >  426.00
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_22 >  235.50
|   |   |   |   |   |   |   |--- feature_22 <= 298.00
|   |   |   |   |   |   |   |   |--- feature_24 <= 113.00
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_24 >  113.00
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 1758.00
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 354.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_1 >  354.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |--- feature_1 >  1758.00
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_22 >  298.00
|   |   |   |   |   |   |   |   |--- feature_24 <= 218.00
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 2.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_25 >  2.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_0 >  1.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_24 >  218.00
|   |   |   |   |   |   |   |   |   |--- feature_23 <= 2.50
|   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_25 >  2.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_23 >  2.50
```

```
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_18 >  0.50
|   |   |   |   |   |   |--- feature_24 <= 476.00
|   |   |   |   |   |   |   |--- feature_24 <= 131.50
|   |   |   |   |   |   |   |   |--- feature_30 <= 4.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_30 >  4.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 1252.21
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1252.21
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_24 >  131.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 204.00
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_22 >  204.00
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 436.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 2517.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  2517.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  436.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |--- feature_24 >  476.00
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |--- feature_5 >  0.50
|   |   |   |   |   |--- feature_24 <= 108.00
|   |   |   |   |   |   |--- feature_30 <= 7.50
|   |   |   |   |   |   |   |--- feature_24 <= 94.50
|   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 1.00
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 364.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_22 >  364.00
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_0 >  1.00
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_24 >  94.50
|   |   |   |   |   |   |   |--- feature_1 <= 4042.50
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_1 >  4042.50
```

```
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |---  feature_30 >  7.50
|   |   |   |   |   |   |---  feature_24 <= 18.50
|   |   |   |   |   |   |   |---  class: False
|   |   |   |   |   |   |---  feature_24 >  18.50
|   |   |   |   |   |   |   |---  class: True
|   |   |   |   |---  feature_24 >  108.00
|   |   |   |   |   |---  feature_24 <= 407.50
|   |   |   |   |   |   |---  feature_24 <= 300.50
|   |   |   |   |   |   |   |---  feature_16 <= 0.50
|   |   |   |   |   |   |   |   |---  feature_1 <= 1380.93
|   |   |   |   |   |   |   |   |   |---  feature_30 <= 5.50
|   |   |   |   |   |   |   |   |   |   |---  feature_22 <= 450.50
|   |   |   |   |   |   |   |   |   |   |   |---  truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |---  feature_22 >  450.50
|   |   |   |   |   |   |   |   |   |   |   |---  class: True
|   |   |   |   |   |   |   |   |   |---  feature_30 >  5.50
|   |   |   |   |   |   |   |   |   |   |---  feature_30 <= 9.50
|   |   |   |   |   |   |   |   |   |   |   |---  truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |---  feature_30 >  9.50
|   |   |   |   |   |   |   |   |   |   |   |---  truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |---  feature_1 >  1380.93
|   |   |   |   |   |   |   |   |   |   |---  feature_25 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |---  feature_1 <= 8745.75
|   |   |   |   |   |   |   |   |   |   |   |   |---  truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |---  feature_1 >  8745.75
|   |   |   |   |   |   |   |   |   |   |   |   |---  class: True
|   |   |   |   |   |   |   |   |   |   |---  feature_25 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |---  feature_23 <= 4.00
|   |   |   |   |   |   |   |   |   |   |   |   |---  truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |---  feature_23 >  4.00
|   |   |   |   |   |   |   |   |   |   |   |   |---  class: False
|   |   |   |   |   |   |   |   |---  feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |---  feature_25 <= 11.50
|   |   |   |   |   |   |   |   |   |   |---  feature_1 <= 2849.50
|   |   |   |   |   |   |   |   |   |   |   |---  feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |---  class: False
|   |   |   |   |   |   |   |   |   |   |   |---  feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |---  truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |---  feature_1 >  2849.50
|   |   |   |   |   |   |   |   |   |   |   |---  feature_18 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |---  class: True
|   |   |   |   |   |   |   |   |   |   |   |---  feature_18 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |---  class: False
|   |   |   |   |   |   |   |   |   |---  feature_25 >  11.50
|   |   |   |   |   |   |   |   |   |   |---  class: True
|   |   |   |   |   |   |---  feature_24 >  300.50
|   |   |   |   |   |   |   |---  feature_29 <= 0.50
```

```
|   |   |   |   |   |   |   |   |--- feature_24 <= 352.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_24 >  352.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |--- feature_24 <= 319.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 8.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  8.50
|   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_25 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- feature_24 >  319.50
|   |   |   |   |   |   |   |   |   |--- feature_21 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_21 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 4.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_23 >  4.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_24 >  407.50
|   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_28 <= 0.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_28 >  0.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |--- class: True
|   |   |--- feature_22 >  466.50
|   |   |   |--- feature_5 <= 0.50
|   |   |   |   |--- feature_2 <= 0.50
|   |   |   |   |   |--- feature_3 <= 0.50
|   |   |   |   |   |   |--- feature_7 <= 0.50
|   |   |   |   |   |   |   |--- feature_8 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_4 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_11 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_10 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_6 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 15
|   |   |   |   |   |   |   |   |   |   |   |--- feature_6 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 16
|   |   |   |   |   |   |   |   |   |--- feature_10 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= -0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 17
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  -0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |--- feature_11 >  0.50
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 691.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 1190.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  1190.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  691.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 606.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  606.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |--- feature_4 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 724.00
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 70.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  70.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 528.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  528.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_22 >  724.00
|   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 8.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  8.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_25 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 1521.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  1521.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_8 >  0.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 758.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= -166.50
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 635.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  635.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_1 >  -166.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 478.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.54
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.54
```

```
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  478.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 15.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  15.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 9
|   |   |   |   |   |   |   |   |--- feature_22 >  758.50
|   |   |   |   |   |   |   |   |   |--- feature_29 <= 3.50
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 933.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  933.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 4.84
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 14
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  4.84
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |--- feature_29 >  3.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 871.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  871.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_27 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_27 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_7 >  0.50
|   |   |   |   |   |   |--- feature_22 <= 730.50
|   |   |   |   |   |   |   |--- feature_24 <= 48.50
|   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_23 <= 11.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 865.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  865.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |--- feature_23 >  11.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 667.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  667.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 2091.50
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |--- feature_1 >  2091.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 4993.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  4993.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_24 >  48.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 110.00
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  110.00
|   |   |   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 214.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  214.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_22 >  730.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 1299.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 66.89
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_26 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_26 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_1 >  66.89
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 478.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  478.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 648.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  648.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 15
|   |   |   |   |   |   |   |   |--- feature_22 >  1299.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 3233.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  4.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_22 >  3233.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |--- feature_3 >  0.50
|   |   |   |   |   |--- feature_22 <= 613.50
|   |   |   |   |   |   |--- feature_25 <= 2.50
|   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 3500.73
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  3500.73
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 477.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  477.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 594.00
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |--- feature_22 >  594.00
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.64
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.64
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_17 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 596.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 589.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 14
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  589.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- feature_22 >  596.50
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 5.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  5.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_17 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 14.59
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  14.59
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |--- feature_25 >  2.50
|   |   |   |   |   |   |   |--- feature_24 <= 266.50
|   |   |   |   |   |   |   |   |--- feature_0 <= 18.00
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_25 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_0 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_0 >  18.00
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_24 >  266.50
|   |   |   |   |   |   |   |   |--- feature_24 <= 343.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_24 >  343.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |--- feature_22 >  613.50
|   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_24 <= 168.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 11.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  11.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_24 >  168.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 1121.50
|   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.03
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |--- feature_0 >  2.03
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 5.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 14
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  5.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_22 >  1121.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 964.92
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  964.92
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  4.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 633.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_22 >  633.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 1521.00
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  1521.00
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |--- feature_2 >  0.50
|   |   |   |   |   |--- feature_24 <= -0.50
|   |   |   |   |   |   |--- feature_22 <= 636.50
|   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1351.21
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1351.21
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 4.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  4.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 606.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.02
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 552.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  552.00
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.02
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.99
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  2.99
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 13
|   |   |   |   |   |   |   |   |   |--- feature_22 >  606.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 625.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 9.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  9.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  625.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 628.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  628.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 506.50
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  506.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 9.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 628.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  628.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  9.50
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 2838.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  2838.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_22 >  636.50
|   |   |   |   |   |   |   |--- feature_29 <= 2.50
|   |   |   |   |   |   |   |   |--- feature_27 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 739.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 1178.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 10
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  1178.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  739.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 14156.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 13
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  14156.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 769.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  769.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |--- feature_27 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 821.00
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 148.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  148.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 1.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  1.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  821.00
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |--- feature_29 >  2.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 790.00
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 709.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  709.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 9.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  9.50
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_26 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_26 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |--- feature_22 >  790.00
|   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.66
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.66
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 507.80
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  507.80
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_24 >  -0.50
|   |   |   |   |   |   |--- feature_24 <= 100.00
|   |   |   |   |   |   |   |--- feature_30 <= 11.50
|   |   |   |   |   |   |   |   |--- feature_29 <= 2.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_29 >  2.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_30 >  1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.78
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.78
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_23 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_30 >  11.50
|   |   |   |   |   |   |   |   |--- feature_0 <= 1.00
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_0 >  1.00
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 635.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  635.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
```

```
|   |   |   |   |   |--- feature_24 >  100.00
|   |   |   |   |   |   |--- feature_0 <= 13.00
|   |   |   |   |   |   |   |--- feature_22 <= 557.00
|   |   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_26 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 9.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 6
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  9.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_26 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 380.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  380.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  557.00
|   |   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 175.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 3142.04
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  3142.04
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  175.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_25 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |   |--- feature_25 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_0 >  13.00
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |--- feature_5 >  0.50
|   |   |   |   |--- feature_22 <= 635.50
|   |   |   |   |   |--- feature_17 <= 0.50
|   |   |   |   |   |   |--- feature_30 <= 2.50
|   |   |   |   |   |   |   |--- feature_22 <= 510.00
|   |   |   |   |   |   |   |   |--- feature_1 <= 203.50
|   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_1 >  203.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 502.00
|   |   |   |   |   |   |   |   |   |--- feature_25 <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_15 >  0.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_25 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.26
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.26
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  502.00
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_22 >  510.00
|   |   |   |   |   |   |   |   |--- feature_1 <= 184.63
|   |   |   |   |   |   |   |   |   |--- feature_25 <= 2.00
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_25 >  2.00
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_1 >  184.63
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 2805.04
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 119.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 542.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  542.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |   |--- feature_24 >  119.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_1 >  2805.04
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_30 >  2.50
|   |   |   |   |   |   |   |--- feature_22 <= 514.50
|   |   |   |   |   |   |   |   |--- feature_24 <= 155.50
|   |   |   |   |   |   |   |   |   |--- feature_29 <= 3.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 227.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 71.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  71.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  227.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |--- feature_29 >  3.50
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_24 >  155.50
|   |   |   |   |   |   |   |   |   |--- feature_23 <= 2.00
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_23 >  2.00
```

```
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_22 >  514.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 531.50
|   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1062.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= -2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  -2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1062.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 5.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_30 >  5.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 3.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  3.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  531.50
|   |   |   |   |   |   |   |   |   |--- feature_23 <= 4.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1888.20
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 1814.30
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 13
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1814.30
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  1888.20
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 3.04
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  3.04
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |--- feature_23 >  4.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 582.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  582.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |--- feature_17 >  0.50
|   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |--- feature_23 <= 5.00
|   |   |   |   |   |   |   |   |--- feature_1 <= -258.38
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_1 >  -258.38
|   |   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_15 >  0.50
```

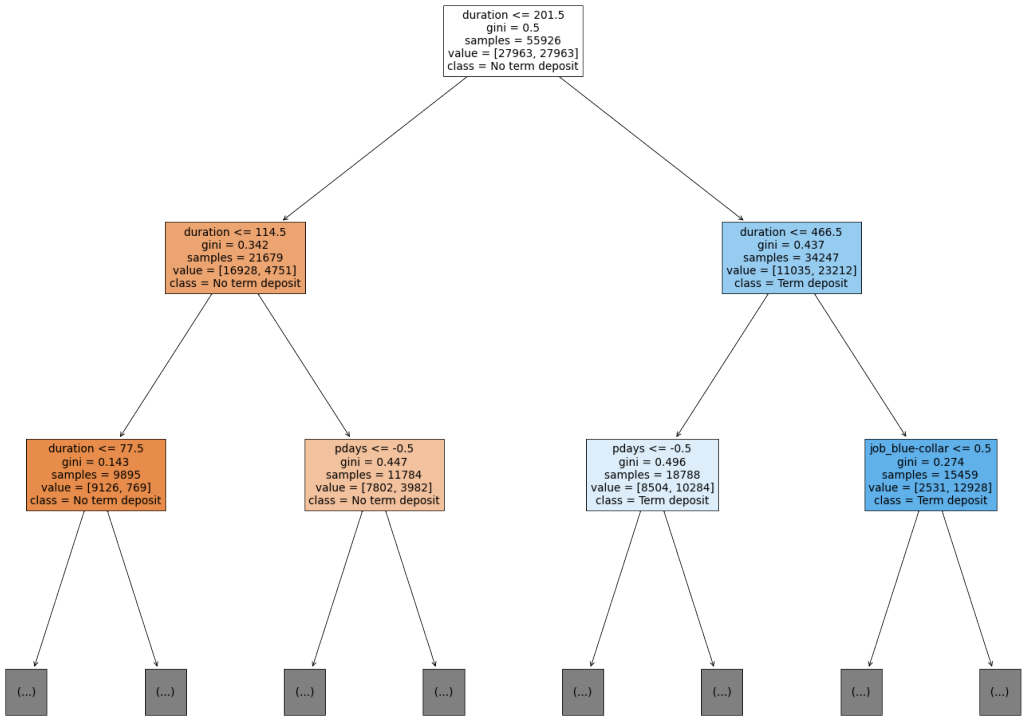
```
|   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_23 >  5.00
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |   |--- class: False
|   |   |   |--- feature_22 >  635.50
|   |   |   |   |--- feature_30 <= 9.50
|   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_15 <= 0.50
|   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_15 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 714.00
|   |   |   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_1 >  714.00
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |--- feature_19 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 16.50
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 319.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= -99.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  -99.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 12
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  319.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 380.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 >  380.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 11
|   |   |   |   |   |   |   |   |   |--- feature_0 >  16.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_19 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_23 <= 4.50
|   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 323.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 1030.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  1030.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_24 >  323.00
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_23 >  4.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 1033.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  1033.50
|   |   |   |   |   |   |   |   |   |   |--- feature_30 <= 5.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_30 >  5.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |--- feature_16 >  0.50
|   |   |   |   |   |   |--- feature_1 <= 4010.50
|   |   |   |   |   |   |   |--- feature_24 <= 339.50
|   |   |   |   |   |   |   |   |--- feature_1 <= -104.50
|   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_1 >  -104.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 1236.00
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 935.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 7
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  935.00
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |   |   |   |   |--- feature_1 >  1236.00
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 684.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_22 >  684.50
|   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 8
|   |   |   |   |   |   |   |--- feature_24 >  339.50
|   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |--- feature_1 >  4010.50
|   |   |   |   |   |   |   |--- class: True
|   |   |   |   |--- feature_30 >  9.50
|   |   |   |   |   |--- feature_22 <= 772.50
|   |   |   |   |   |   |--- feature_30 <= 10.50
|   |   |   |   |   |   |   |--- feature_1 <= 245.00
|   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |--- feature_1 >  245.00
|   |   |   |   |   |   |   |   |--- feature_0 <= 2.50
|   |   |   |   |   |   |   |   |   |--- feature_16 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_20 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_20 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_16 >  0.50
```

```
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_0 >  2.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_30 >  10.50
|   |   |   |   |   |   |   |   |--- feature_13 <= 0.50
|   |   |   |   |   |   |   |   |   |--- feature_23 <= 3.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_23 >  3.50
|   |   |   |   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_13 >  0.50
|   |   |   |   |   |   |   |   |   |--- feature_30 <= 11.50
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |--- feature_30 >  11.50
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |--- feature_22 >  772.50
|   |   |   |   |   |   |   |--- feature_14 <= 0.50
|   |   |   |   |   |   |   |   |--- feature_30 <= 11.50
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.01
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 838.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  838.00
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 980.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  980.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |--- feature_0 >  2.01
|   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |--- feature_30 >  11.50
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 1069.00
|   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  1069.00
|   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 506.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  506.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_1 <= 2827.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_1 >  2827.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |--- feature_14 >  0.50
|   |   |   |   |   |   |   |   |--- feature_22 <= 997.00
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 849.00
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 797.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 2.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_29 >  2.50
```

```
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  797.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  849.00
|   |   |   |   |   |   |   |   |   |   |--- feature_27 <= 0.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_27 >  0.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 913.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_22 >  913.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |--- feature_22 >  997.00
|   |   |   |   |   |   |   |   |   |--- feature_22 <= 1307.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 5.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  5.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 11.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  11.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_22 >  1307.50
|   |   |   |   |   |   |   |   |   |   |--- feature_29 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 <= 158.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |   |--- feature_24 >  158.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: False
|   |   |   |   |   |   |   |   |   |--- feature_29 >  1.50
|   |   |   |   |   |   |   |   |   |   |--- feature_22 <= 1339.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: True
|   |   |   |   |   |   |   |   |   |   |--- feature_22 >  1339.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: False
```

[37]:
```
fig = plt.figure(figsize = (25, 20))
_ = tree.plot_tree(clf_tree,
                   feature_names = list(X_train.columns),
                   class_names = ['No term deposit', 'Term deposit'],
                   max_depth = 2,
                   filled = True)
```



27 Train a random forest model

[38]:
```
rf = RandomForestClassifier()
rf.fit(X_train_SMOTE, y_train_SMOTE)
```

[38]:
```
RandomForestClassifier()
```

28 Make predictions using the random forest model

[39]:
```
rf_y_pred = rf.predict(X_test)
print(Counter(rf_y_pred))
```

```
Counter({False: 12488, True: 1066})
```

[40]:
```
results['term_deposit_pred'] = rf_y_pred[:5]
results
```

[40]:

	term_deposit	term_deposit_pred	number_transactions	total_amount_usd	\
43308	False	False	0.0	1369.42	
32770	False	False	2.0	246.00	
17440	False	True	0.0	1369.42	
36164	False	False	0.0	1369.42	
29218	False	False	0.0	1369.42	

	job_management	job_technician	job_entrepreneur	job_blue-collar	\
43308	1	0	0	0	
32770	0	0	0	1	
17440	0	0	0	0	
36164	0	0	0	1	
29218	0	0	0	0	

	job_retired	job_admin.	…	contact_telephone	duration	campaign	\
43308	0	0	…	1	129	1	
32770	0	0	…	0	191	1	
17440	0	0	…	0	780	2	
36164	0	0	…	0	114	2	
29218	0	0	…	0	111	1	

	pdays	previous	device_desktop	device_tablet	single	\
43308	188	2	0	0	0	
32770	150	7	0	0	0	
17440	-1	0	0	0	0	
36164	-1	0	1	0	1	
29218	-1	0	0	0	0	

	age_group_encoded	month_joined
43308	2	3
32770	1	7
17440	2	5
36164	2	10
29218	2	5

[5 rows x 33 columns]

29 Obtain the random forest model's score

[41]:
```
accuracy_score(y_test, rf_y_pred)
```

[41]: 0.8826176774383946

30 Train a gradient boosting model

[42]:
```
xgb = XGBClassifier(eval_metric = 'logloss', n_jobs = 1)
xgb.fit(X_train_SMOTE, y_train_SMOTE)
```

[42]:
```
XGBClassifier(base_score=0.5, booster='gbtree', colsample_bylevel=1,
              colsample_bynode=1, colsample_bytree=1, eval_metric='logloss',
              gamma=0, gpu_id=-1, importance_type='gain',
              interaction_constraints='', learning_rate=0.300000012,
              max_delta_step=0, max_depth=6, min_child_weight=1, missing=nan,
              monotone_constraints='()', n_estimators=100, n_jobs=1,
              num_parallel_tree=1, random_state=0, reg_alpha=0, reg_lambda=1,
              scale_pos_weight=1, subsample=1, tree_method='exact',
              validate_parameters=1, verbosity=None)
```

31 Make predictions using the gradient boosting model

[43]:
```
xgb_y_pred = xgb.predict(X_test)
print(Counter(xgb_y_pred))
```

```
Counter({False: 12287, True: 1267})
```

[44]:
```
results['term_deposit_pred'] = xgb_y_pred[:5]
results
```

[44]:

	term_deposit	term_deposit_pred	number_transactions	total_amount_usd	\
43308	False	False	0.0	1369.42	
32770	False	False	2.0	246.00	
17440	False	True	0.0	1369.42	
36164	False	False	0.0	1369.42	
29218	False	False	0.0	1369.42	

	job_management	job_technician	job_entrepreneur	job_blue-collar	\
43308	1	0	0	0	
32770	0	0	0	1	
17440	0	0	0	0	
36164	0	0	0	1	
29218	0	0	0	0	

	job_retired	job_admin.	…	contact_telephone	duration	campaign	\
43308	0	0	…	1	129	1	
32770	0	0	…	0	191	1	
17440	0	0	…	0	780	2	
36164	0	0	…	0	114	2	
29218	0	0	…	0	111	1	

```
       pdays  previous  device_desktop  device_tablet  single  \
43308    188         2               0              0       0
32770    150         7               0              0       0
17440     -1         0               0              0       0
36164     -1         0               1              0       1
29218     -1         0               0              0       0

       age_group_encoded  month_joined
43308                  2             3
32770                  1             7
17440                  2             5
36164                  2            10
29218                  2             5

[5 rows x 33 columns]
```

32 Obtain the gradient boosting model's score

[45]:
```
accuracy_score(y_test, xgb_y_pred)
```

[45]:
```
0.8854950568097979
```

33 Define the parameter grid used to tune the logistic regression model

[46]:
```
solvers = ['newton-cg', 'lbfgs', 'liblinear']
penalty = ['l2']
c_values = [10, 1.0, 0.1, 0.01]

param_grid = dict(solver = solvers, penalty = penalty, C = c_values)

print(param_grid)
```

```
{'solver': ['newton-cg', 'lbfgs', 'liblinear'], 'penalty': ['l2'], 'C': [10,
1.0, 0.1, 0.01]}
```

34 Perform a randomized search for optimal hyperparameters

[47]:
```
model = LogisticRegression()
random_search = RandomizedSearchCV(estimator = model,
                                   param_distributions = param_grid)
random_search.fit(X_train_norm, y_train_SMOTE)
```

[47]:
```
RandomizedSearchCV(estimator=LogisticRegression(),
                   param_distributions={'C': [10, 1.0, 0.1, 0.01],
                                        'penalty': ['l2'],
                                        'solver': ['newton-cg', 'lbfgs',
                                                   'liblinear']})
```

[48]:
```
# Summarize the results of the randomized search.

print('Best accuracy score:', round(random_search.best_score_, 4))
print('Best parameters:    ', random_search.best_params_)
```

```
Best accuracy score: 0.8949
Best parameters:     {'solver': 'newton-cg', 'penalty': 'l2', 'C': 1.0}
```

35 Perform a grid search for optimal hyperparameters

[49]:
```
model = LogisticRegression()
grid_search = GridSearchCV(estimator = model,
                           param_grid = param_grid)
logreg_fit = grid_search.fit(X_train_norm, y_train_SMOTE)
```

[50]:
```
# Summarize the results of the grid search.

print('Best accuracy score:', round(grid_search.best_score_, 4))
print('Best parameters:    ', grid_search.best_params_)
```

```
Best accuracy score: 0.8949
Best parameters:     {'C': 1.0, 'penalty': 'l2', 'solver': 'newton-cg'}
```

36 Tune the gradient boosting model to reduce overfitting

[51]:
```
model = XGBClassifier(eval_metric = 'logloss', n_jobs = 1)

param_grid = {
    'n_estimators': [10],
    'max_depth': [15, 25],
```

```
    'reg_alpha': [1.1, 1.2],
    'reg_lambda': [1.2, 1.3]
}
```

[52]:
```
gs = GridSearchCV(estimator = model,
                  param_grid = param_grid,
                  n_jobs = 1,
                  scoring = 'accuracy',
                  verbose = 2)

fitted_model = gs.fit(X_train_SMOTE, y_train_SMOTE)
```

```
Fitting 5 folds for each of 8 candidates, totalling 40 fits
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …

[Parallel(n_jobs=1)]: Using backend SequentialBackend with 1 concurrent workers.

[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
2.3s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …

[Parallel(n_jobs=1)]: Done   1 out of   1 | elapsed:    2.3s remaining:    0.0s

[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
2.6s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
2.2s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
2.6s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
2.5s
```

```
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
2.3s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
2.1s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
2.5s
[CV] max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=15, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
2.5s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
2.4s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.2, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
2.4s
```

```
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
3.1s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.1, reg_lambda=1.3, total=
3.1s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
2.5s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
3.1s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.2, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
2.5s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
3.1s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
3.0s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
3.1s
[CV] max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3 …
[CV]  max_depth=25, n_estimators=10, reg_alpha=1.2, reg_lambda=1.3, total=
3.1s

[Parallel(n_jobs=1)]: Done  40 out of  40 | elapsed:  1.8min finished
```

[53]:
```
print('Best accuracy score:', round(gs.best_score_, 4))
print('Best parameters:    ', gs.best_params_)
```

```
Best accuracy score: 0.9092
Best parameters:     {'max_depth': 25, 'n_estimators': 10, 'reg_alpha': 1.1,
'reg_lambda': 1.3}
```

37 Compare evaluation metrics for each model

[54]:
```
models = ['Logistic Regression', 'Naïve Bayes', 'SVM', 'k-NN',
          'Decision Tree', 'Random Forest', 'XGBoost', 'Dummy Classifier']

metrics = ['Accuracy', 'Precision', 'Recall', 'F1']

pred_list = ['logreg_y_pred', 'gnb_y_pred', 'svm_y_pred', 'knn_y_pred',
             'clf_tree_y_pred','rf_y_pred', 'xgb_y_pred', 'dummy_y_pred']

# Baseline algorithm.
dummy = DummyClassifier(strategy = 'stratified')
dummy.fit(X_train_SMOTE, y_train_SMOTE)
dummy_y_pred = dummy.predict(X_test)

scores = np.empty((0, 4))

for i in pred_list:
    scores = np.append(scores,
                       np.array([[accuracy_score(y_test, globals()[i]),
                                  precision_score(y_test, globals()[i]),
                                  recall_score(y_test, globals()[i]),
                                  f1_score(y_test, globals()[i])]]),
                       axis = 0)

scores = np.around(scores, 4)

scoring_df = pd.DataFrame(scores, index = models, columns = metrics)
scoring_df.sort_values(by = 'F1', ascending = False)
```

[54]:

	Accuracy	Precision	Recall	F1
XGBoost	0.8855	0.5280	0.4122	0.4630
Random Forest	0.8826	0.5150	0.3383	0.4083
Decision Tree	0.8314	0.3309	0.3993	0.3619
Naïve Bayes	0.7019	0.1876	0.4473	0.2644
Logistic Regression	0.1248	0.1203	1.0000	0.2148
k-NN	0.2969	0.1123	0.7055	0.1937
Dummy Classifier	0.4964	0.1171	0.4904	0.1891
SVM	0.8803	0.0000	0.0000	0.0000

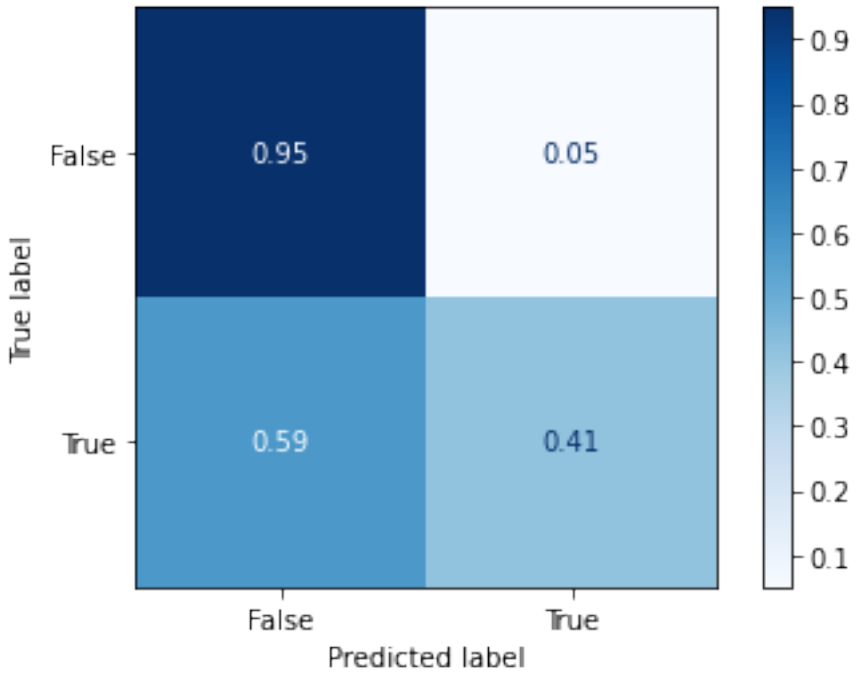
38 Generate a confusion matrix

[55]:
```
confusion_matrix(y_test, xgb_y_pred)
```

[55]:
```
array([[11333,   598],
       [  954,   669]])
```

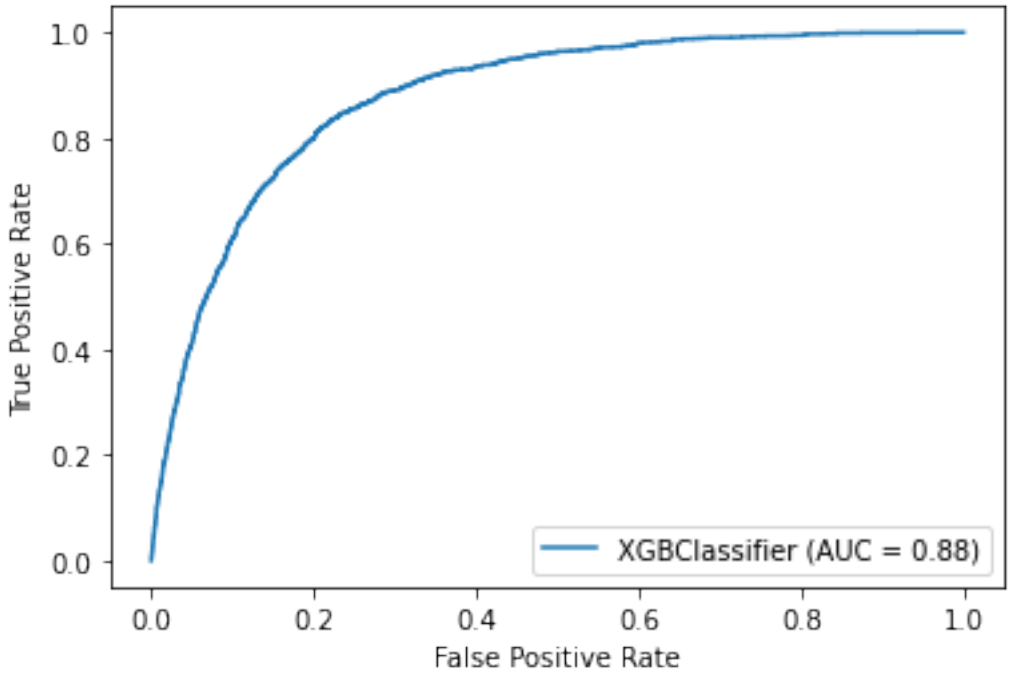
[56]:
```
plot_confusion_matrix(xgb,
                      X_test,
                      y_test,
                      cmap = plt.cm.Blues,
                      normalize = 'true')

plt.show();
```



39 Plot a ROC curve

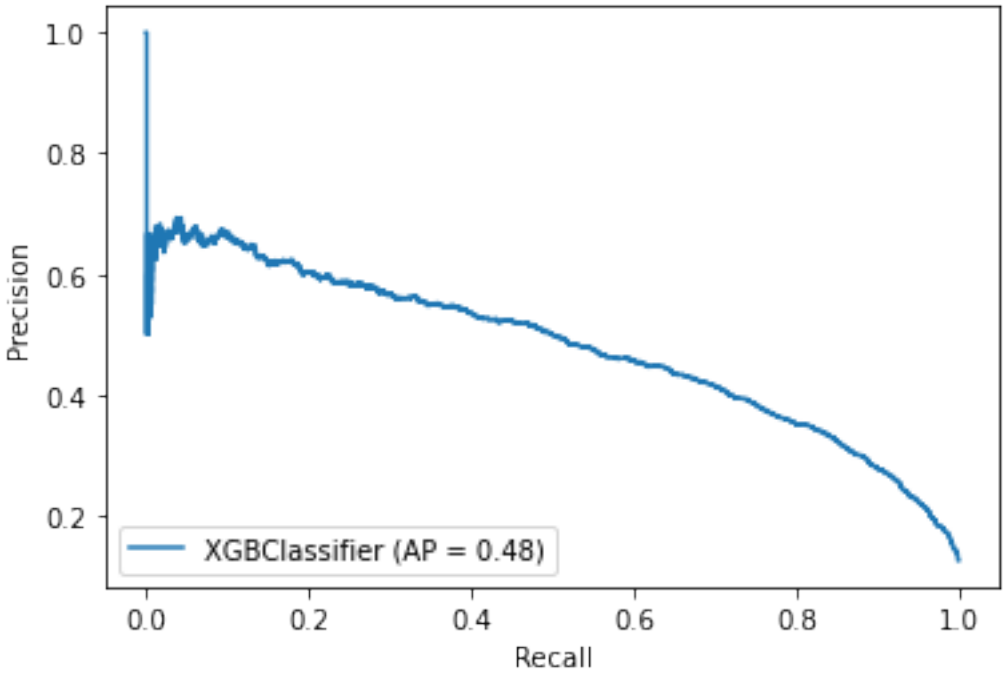
[57]:
```
plot_roc_curve(xgb, X_test, y_test)
plt.show()
```



40 Plot a precision–recall curve

[58]:
```
plot_precision_recall_curve(xgb, X_test, y_test)

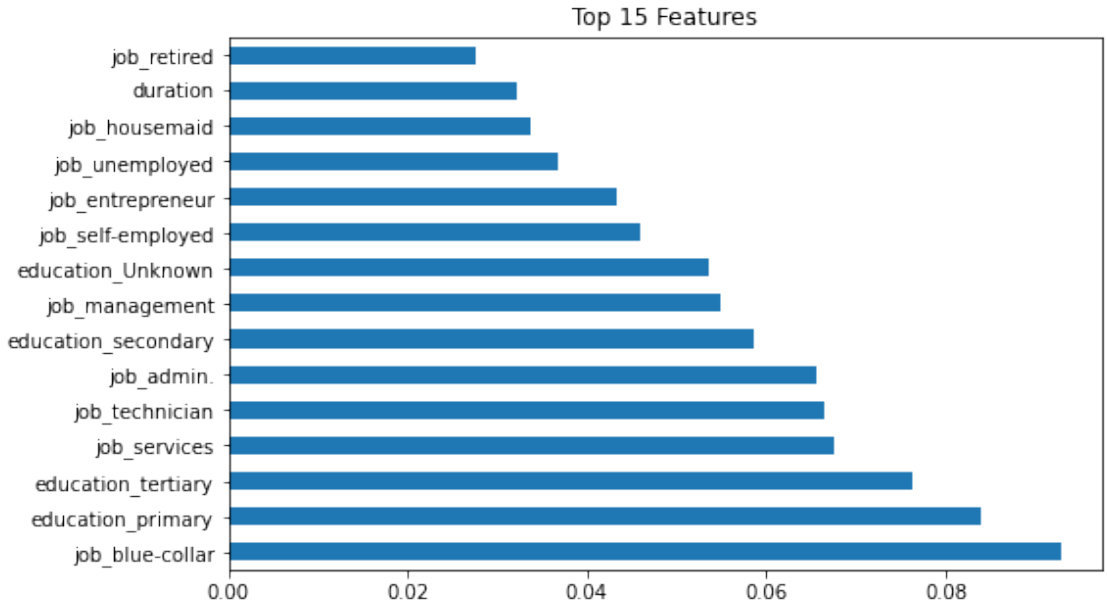
plt.show();
```



41 Generate a feature importance plot

[59]:
```
def feature_importance_plot(model, X_train, n):
    """Plots feature importance. Only works for ensemble learning."""
    plt.figure(figsize = (8, 5))
    feat_importances = pd.Series(model.feature_importances_,
                                 index = X_train.columns)
    feat_importances.nlargest(n).plot(kind = 'barh')
    plt.title(f'Top {n} Features')
    plt.show()
```

[60]:
```
feature_importance_plot(xgb, X_train_SMOTE, 15)
```



42 Plot learning curves

[1]:
```
def plot_learning_curves(model, X_train, y_train):
    """Plots learning curves for model validation."""
    plt.figure(figsize = (5, 5))
    train_sizes, train_scores, test_scores = \
    learning_curve(model, X_train, y_train, cv = 5,
                   scoring = 'accuracy', n_jobs = 1,
                   shuffle = True,
                   train_sizes = np.linspace(0.01, 1.0, 5))

    # Means of training and test set scores.
    train_mean = np.mean(train_scores, axis = 1)
    test_mean = np.mean(test_scores, axis = 1)

    # Draw lines.
    plt.plot(train_sizes, train_mean, '--',
             color = '#111111', label = 'Training score')
    plt.plot(train_sizes, test_mean,
             color = '#111111', label = 'Cross-validation score')

    # Create plot.
    plt.title('Learning Curves')
    plt.xlabel('Training Set Size')
    plt.ylabel('Accuracy Score')
```

```
    plt.legend(loc = 'best')
    plt.tight_layout()

    plt.show()
```

[2]:
```
plot_learning_curves(xgb, X_train_SMOTE, y_train_SMOTE)
```

```
      ␣
↪---------------------------------------------------------------------------

       NameError                                 Traceback (most recent call␣
↪last)

       <ipython-input-2-53a58766d913> in <module>
   ----> 1 plot_learning_curves(xgb, X_train_SMOTE, y_train_SMOTE)


       NameError: name 'xgb' is not defined
```

43 Save the best model

[ ]:
```
pickle.dump(xgb, open('xgboost_classifier.pickle', 'wb'))
```